

MAJOR PROJECT REGISTRATION FORM

Date:

Project Category:	☐ Research oriented	☐ Application oriented	Batch ID:	06
	☐ Product oriented	☐ Other (please specify)		
Specify if others:				

Title of the Project: AI-Based Research Paper Novelty Assessment System Using Retrieval-Augmented Generation (RAG)

Abstract

The rapid growth of scientific publications has made it increasingly difficult for researchers to evaluate the originality and novelty of new research papers through manual literature review. Existing plagiarism detection tools primarily identify copied text but do not assess conceptual novelty or highlight potential research gaps. This project proposes an **AI-Based Research Paper Novelty Assessment System using Retrieval-Augmented Generation (RAG)** to assist researchers in evaluating the originality of their work.

The proposed system allows users to upload a research paper in PDF format. The paper is processed to extract textual content, and semantic embeddings are generated using transformer-based language models. A RAG framework retrieves the most relevant research papers from a vector database through semantic similarity search. Based on the retrieved literature, the system analyzes similarities and differences in objectives, methodologies, datasets, and findings. It then provides an estimated novelty assessment, highlights possible research gaps, and generates an AI-assisted analytical report to support researchers during the literature review process.

The proposed system is designed as an **AI-powered decision-support tool** rather than a replacement for expert evaluation. By integrating Retrieval-Augmented Generation (RAG), Natural Language Processing (NLP), transformer-based embeddings, and semantic retrieval techniques, it simplifies literature review, assists researchers in assessing research novelty, identifies possible research gaps, and supports the development of more innovative and impactful research.

Overall, the proposed system provides an intelligent and efficient framework for research paper novelty assessment by leveraging Retrieval-Augmented Generation (RAG) and semantic analysis. It assists researchers in identifying similar studies, understanding existing research trends, and exploring possible research gaps, thereby promoting faster, more effective, and innovation-driven research.

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